

BIA-810 Final Project Report

Healthcare Analytics

Analyzing Market Dynamics & Revitalizing Brand Strategy for Market Cannibalization of an Injectable Anesthesia Drug

A data-driven analysis of the U.S. injectable anesthesia market (2016–2018), examining why the planned cannibalization between Product 1 (J1885 — ketorolac) and Product 2 (J2250 — midazolam) failed, and what strategic actions are required to recover Product 2's market share against the rapidly growing competitor Product 3 (J3010 — fentanyl).

Submitted: May 12, 2025

*Data Source: Medicare CCLF Claims (2016–2018); HCP Demographics; Patient Demographics; Zip-to-Territory Mapping;
ICD-10 Diagnosis Specialty Mapping*

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1. Executive Summary

The cannibalization play has failed. Product 2 (J2250 — midazolam) was launched to absorb the dropping share of Product 1 (J1885 — ketorolac). Instead, Product 3 (J3010 — fentanyl) — a competitor — has captured almost all of that share and has now overtaken Product 1 in revenue terms. Product 2 has been quietly compressed into clinical irrelevance.

Three findings that explain everything

1. Wrong substitute. Midazolam (J2250) is a benzodiazepine sedative; ketorolac (J1885) is an NSAID for pain. They are not in the same drug class and were never clinically substitutable. The cannibalization assumption was a portfolio-level error.
2. Retention crisis, not acquisition crisis. J2250 loses more than half its writers every year (46.4% retention) and has zero High-Volume prescribers. No level of new-writer recruitment can outrun this churn.
3. Direct competitive displacement. 58% of J2250 patients have also received J3010. The competitor is not just winning new patients — it is treating our patients.

Key headline metrics

Metric	Value	Implication
J2250 Claims CAGR (2016–18)	–16.1%	Brand is in structural decline
J3010 Claims CAGR (2016–18)	+62.7%	Competitor expanding 5x our rate
J2250 Writer Retention (2017→18)	46.4%	Half the base churns every year
J3010 Writer Retention (2017→18)	94.5%	Competitor's loyalty compounding
J2250 Patients also on J3010	58%	Direct displacement, not greenfield
Territories in Decline (J2250)	15 of 22	Geographic erosion is broad
J3010 2018 Revenue Share	44.4%	J3010 has overtaken J1885 (\$)

Top three sized actions

4. Fix retention first. Re-engage the lapsed J2250 writers from 2018 (~148 NPIs). Lift retention 46% → 65% in 12 months → +120 to +200 incremental claims.
5. Convert Trialists into Rising Stars. 90% of J2250 writers sit at 1–4 claims/year. Doubling claim volume for just 20% of the 284 Trialists adds ~+170 claims with zero acquisition cost.
6. Declare 4 emergency territories. St. Louis, Phoenix, LA–San Diego, New York. Together they lost 70+ J2250 claims while J3010 gained 165+ in the same markets in a single year.

2. Project Background & Problem Statement

2.1 Market context

The injectable anesthesia market is a sub-segment of the pharmaceutical industry covering medications used to induce and maintain anesthesia during medical procedures and surgeries. Market demand is tightly coupled to the volume of surgical procedures, which continues to grow due to an aging population, expanding healthcare access, and ongoing advances in surgical technology.

2.2 The four products in scope

HCPCS	Generic Name	Role	Description
J1885	Ketorolac tromethamine	Product 1 — Leader (declining)	NSAID; short-term moderate to severe pain (pre/post-op)
J2250	Midazolam HCl	Product 2 — Variant (our brand)	Benzodiazepine; induction of general anesthesia / sedation
J3010	Fentanyl citrate	Product 3 — Main competitor	Synthetic opioid; analgesia during anesthesia, induction, recovery
J2704	Propofol	Product 4 — Alternative competitor	IV anesthetic / sedative; sedation and induction

2.3 The problem

Our company is a pharmaceutical leader in the injectable anesthesia portfolio. Our market-leading brand Product 1 (J1885) has experienced consistent share decline. Product 2 (J2250) was recently launched as a portfolio variant designed to cannibalize Product 1 and capture the dropping share ourselves. The cannibalization is not happening — Product 3 (J3010) is rapidly gaining share, while Product 2 continues to lose ground.

Our task is to use the Medicare CCLF claims data to (i) explain the failure, (ii) identify the key market drivers, and (iii) recommend evidence-based, sized actions to recover Product 2's share.

3. Methodology & Data Preparation

3.1 Data lineage

Five compressed Medicare CCLF claim files were unzipped and merged with HCP demographics, patient demographics, ZIP-to-territory mapping, and ICD-10 specialty mapping into a single analysis-ready dataset of 28,368 line items belonging to claims that include any of the four market-basket HCPCS codes (J1885, J2250, J2704, J3010).

3.2 Key analytical decisions

Decision	Choice / Value	Rationale
Time window	2016–2018	Per project brief. 17 line items from 2015 dropped (out of window).
Brand attribution	Line-level <code>clm_line_hcpcs_cd</code> $\in$ market basket (15,262 brand line items, 15,139 unique claim IDs)	Each line item identifies a specific drug administration; aggregating to claim-ID would risk double-counting administrations of multiple basket products on the same encounter.
Date field	<code>clm_from_dt</code> parsed via <code>pd.to_datetime</code> ; <code>clm_thru_dt</code> normalized from mixed ISO/US formats	Consistent with project guidance.
ZIP code handling	Cast to 5-digit zero-padded string before territory join	Preserves leading zeros (e.g., 01104) lost during CSV import.
Diagnosis specialty	Initial alphabet of <code>clm_dgns_cd</code> mapped via Diagnosis Code Mapping	Per assignment hints. Principal diagnosis (<code>prncpl_dgns_cd</code>) reviewed as supplementary.
HCP definition	Unique <code>fac_prvdr_npi_num</code> (10-digit NPI)	Standard claims convention.
New writer	NPI's first observed claim year for that brand	Subject to a 2016 boundary effect (data start year) — flagged on Chart 12.
Continuing writer	NPI active in both years of the period	Per assignment.
Revenue / spend metric	<code>clm_line_cvrpd_pd_amt</code> (Medicare-paid amount)	Cleanest available proxy for brand revenue. Medicare-only — not commercial book.

3.3 Data quality checks

- Total dataset: 28,368 rows verified via assertion.
- Records excluded: 17 from 2015, kept silent in original output but explicitly noted here.
- Brand line items in scope: 15,262 across 15,139 unique claim IDs.
- Share columns sum to ~100% per year (within float tolerance) — verified.
- Date format inconsistencies in `clm_thru_dt` resolved by re-parsing with `pd.to_datetime`.
- Upstream territory typos preserved as-is (e.g., 'Philedelphia', 'Pittsburg') — flagged in limitations.

4. Strategic Reframe — Why Cannibalization Was Always Going to Fail

The case has been framed as a market-execution problem. The data shows it was structurally a pharmacological problem.

HCPCS	Generic	Drug Class	Clinical Role
J1885	Ketorolac tromethamine	NSAID	Short-term moderate-to-severe pain; pre/post-op analgesic
J2250	Midazolam HCl	Benzodiazepine	Induction of general anesthesia / sedation
J3010	Fentanyl citrate	Synthetic opioid	Analgesia during anesthesia, induction, recovery
J2704	Propofol	IV anesthetic / sedative	Sedation / induction

INSIGHT. J2250 (a sedative) and J1885 (an NSAID) are not clinically substitutable. They sit at different points in the perioperative pathway. The 'cannibalization plan' assumed they were portfolio substitutes when in fact J2250's natural competitive set is J2704 (propofol) — both are induction/sedation agents — and, in mixed sedation–analgesia protocols, J3010 (fentanyl).

STRATEGIC IMPLICATION. The brand strategy must be reset against the real competitive set. Detail aids, KOL programs, and clinical evidence generation should reposition J2250 in the induction / sedation conversation against J2704 and J3010 — not against J1885. This single reframe is the most important strategic move available.

5. Q1 — Market Dynamics & Competitive Landscape

Comprehensive analysis of the injectable anesthesia market for 2016–2018: trends in market share, growth rates, and key indicators for Products 1, 2, 3, and 4. We examine claims, patients, and HCPs (writers); claims-per-HCP and patients-per-HCP productivity; and the territory-level dynamics of the variant brand's decline against the main competitor's rise.

5.1 Chart 1 — Claims Share by Product (100% Stacked)

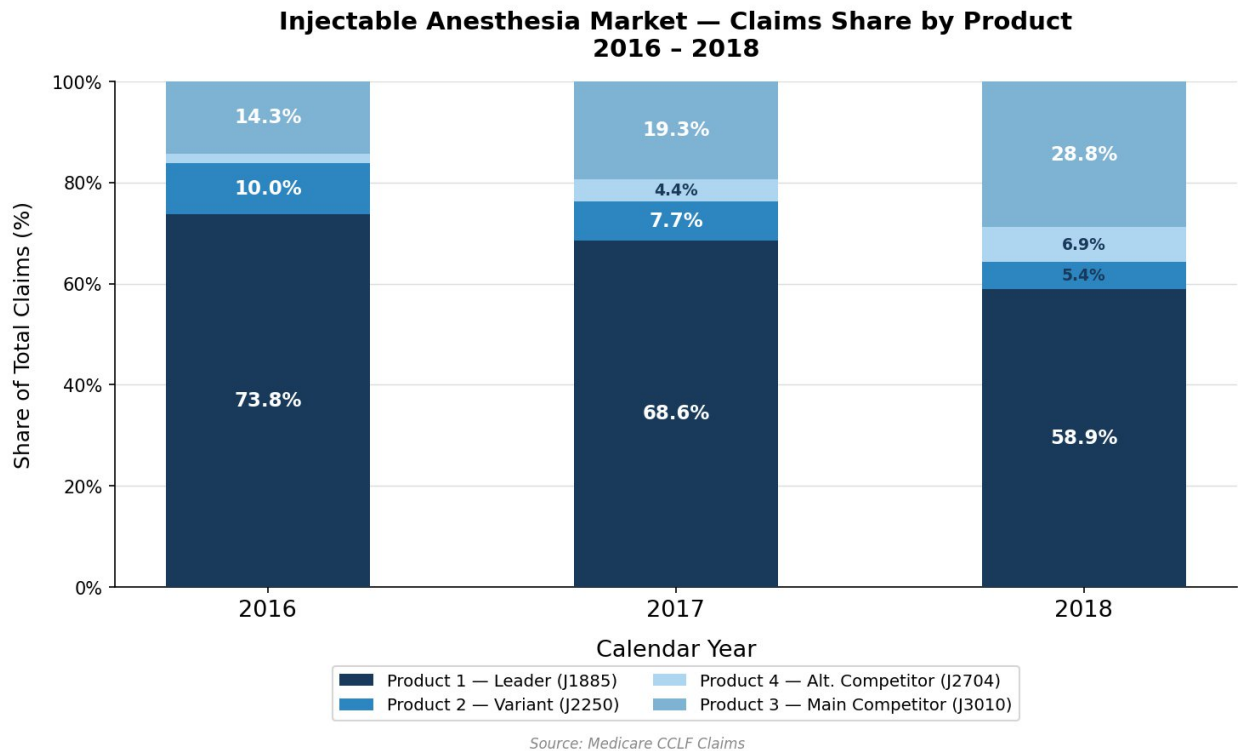


Chart 1: Share of total injectable anesthesia claims by product, 2016–2018.

What it shows.

Product 1 (Leader) held 73.8% of all claims in 2016 but dropped to 58.9% by 2018. Product 3 (Main Competitor) grew from 14.3% to 28.8%. Product 2 (our Variant) barely moved — from 10.0% to just 5.4%, meaning it lost share rather than absorbing Product 1's drop.

INSIGHT. The cannibalization strategy has failed completely. Every percentage point that Product 1 loses is going to the competitor, not our own brand.

RECOMMENDATION. Product 2 needs an urgent commercial reset before Product 3 becomes the default alternative. Reset the go-to-market — messaging, targeting, and channel mix — anchored on the pharmacological reframe in Section 4.

5.2 Chart 2 — Unique Patient Share by Product

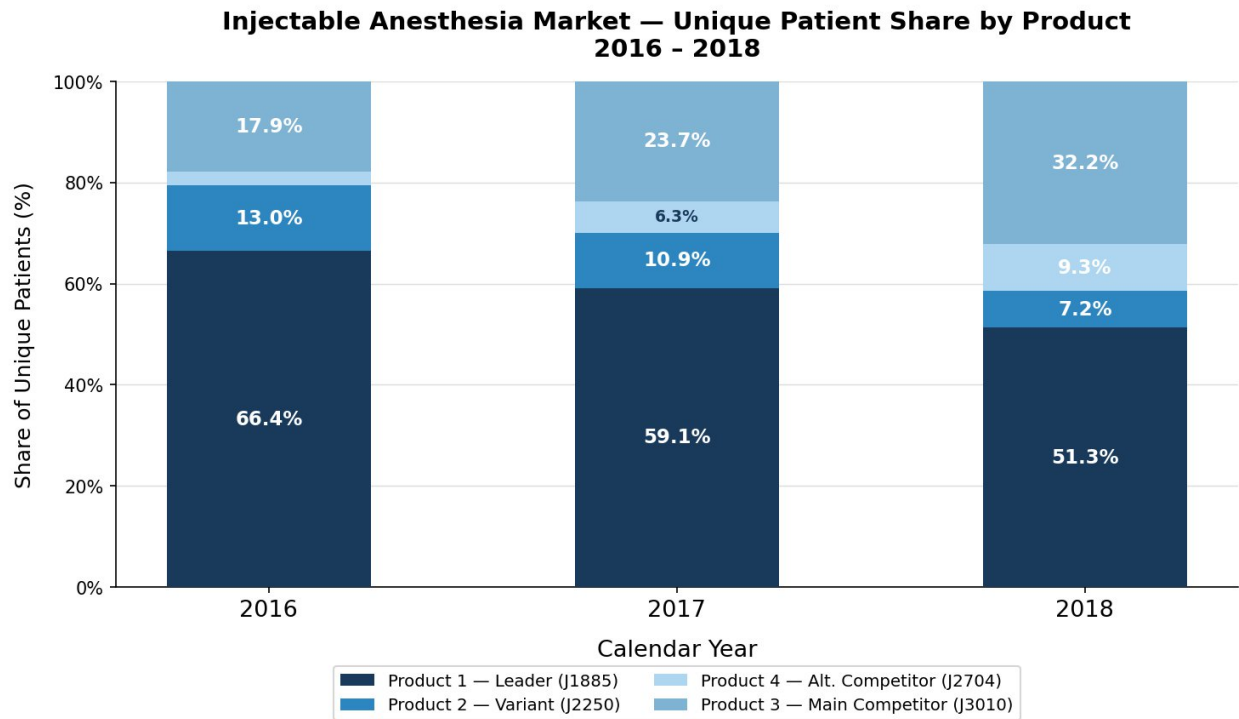


Chart 2: Share of unique patients by product, 2016–2018.

What it shows.

Product 1's patient share fell from 66.4% to 51.3%. Product 3's patient share grew from 17.9% to 32.2%. Product 2's patient share fell from 13.0% to 7.2%.

INSIGHT. Product 3 is not just winning new patients — it is treating patients who previously received Product 1 and Product 2. At the current rate, Product 3 will surpass Product 1 in patient share within 1–2 years.

RECOMMENDATION. Patient retention programs must be launched immediately. HCPs who currently treat patients with Product 2 should receive personalised outreach focused on the clinical benefits of staying on the brand. Switching patients back once they have moved to the competitor is far harder than retaining them now.

5.3 Chart 3 — Unique HCP (Writer) Share by Product

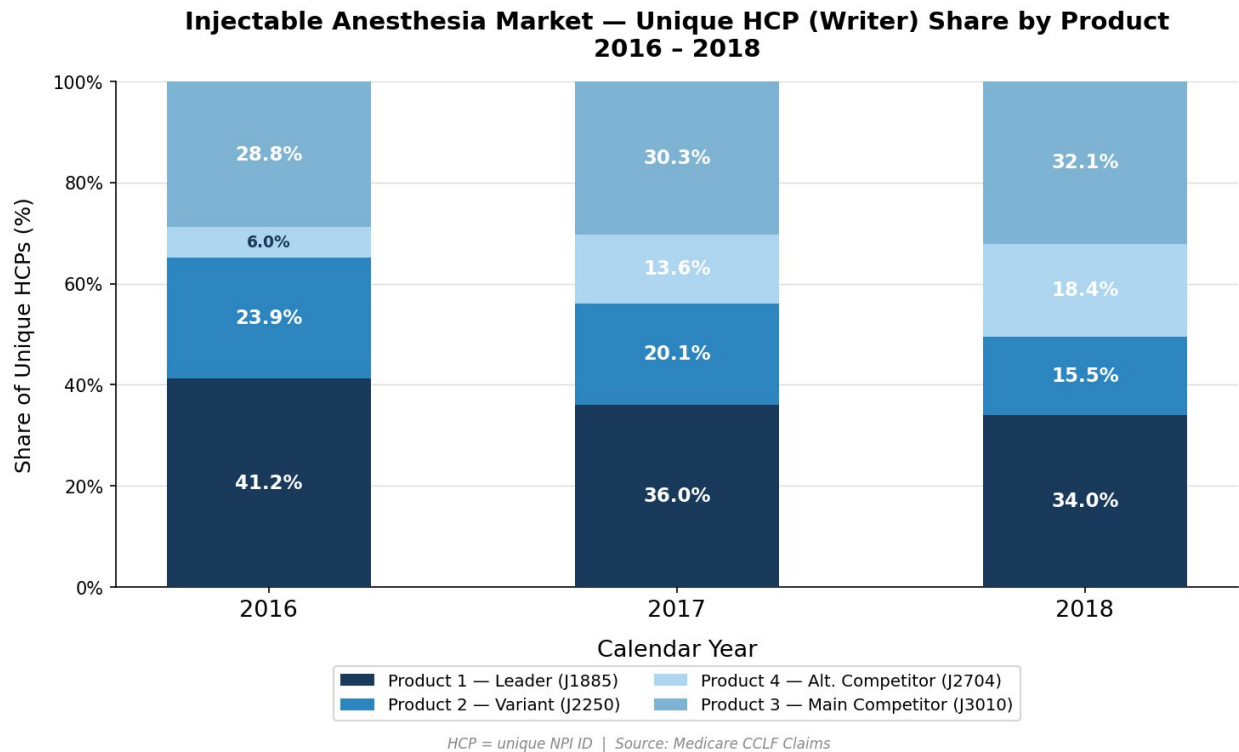


Chart 3: Share of unique HCP writers by product, 2016–2018.

What it shows.

Product 2's HCP writer share fell from 23.9% to 15.5%. Product 3's writer share grew from 28.8% to 32.1%. Product 1 dropped from 41.2% to 34.0%.

INSIGHT. Product 2 is losing HCPs faster than it is losing patients. This is a leading indicator — when doctors stop writing a brand, patients follow shortly after. The HCP writer base is the foundation of any brand's commercial performance, and Product 2's foundation is collapsing.

RECOMMENDATION. Identify and prioritize the HCPs who switched away from Product 2 in 2018. Deploy the sales force to re-engage these specific NPIs with a targeted clinical message.

5.4 Chart 4 — Claims per HCP (Writer Productivity)

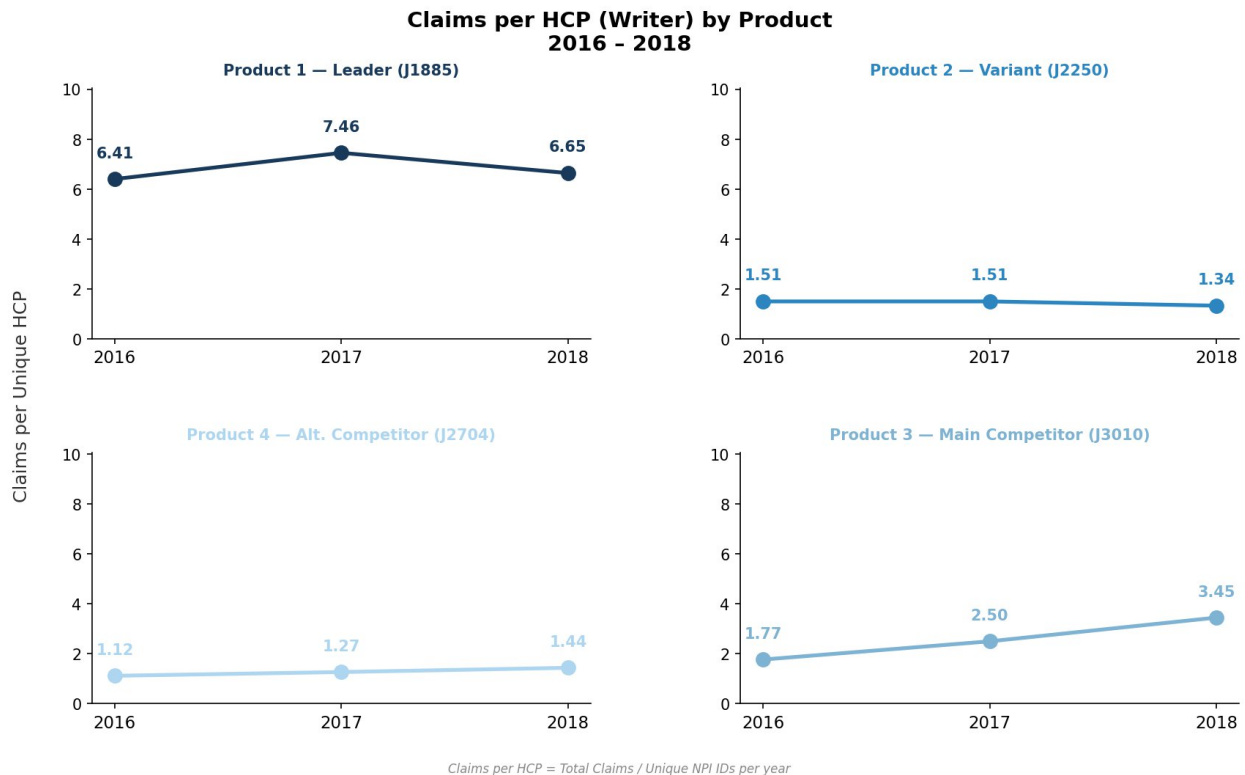


Chart 4: Annual claims per HCP, by product, 2016–2018.

What it shows.

Product 1 writers average 6.41–7.46 claims per HCP. Product 3 writers grew from 1.77 to 3.45 claims per HCP. Product 2 writers average only 1.34–1.51 — barely above one claim per writer per year.

INSIGHT. Product 2 writers are not committed. They write the brand once or twice and stop. Product 3 writers are becoming increasingly productive — each one is writing more claims every year, which means deeper clinical adoption.

RECOMMENDATION. Focus sales effort on converting existing Product 2 Trialist writers (2–4 claims) into Rising Star writers (5–9 claims). A rep-delivered clinical case-study program or a peer speaker event featuring a high-volume Product 2 writer is the highest-leverage move available.

5.5 Chart 5 — Patients per HCP

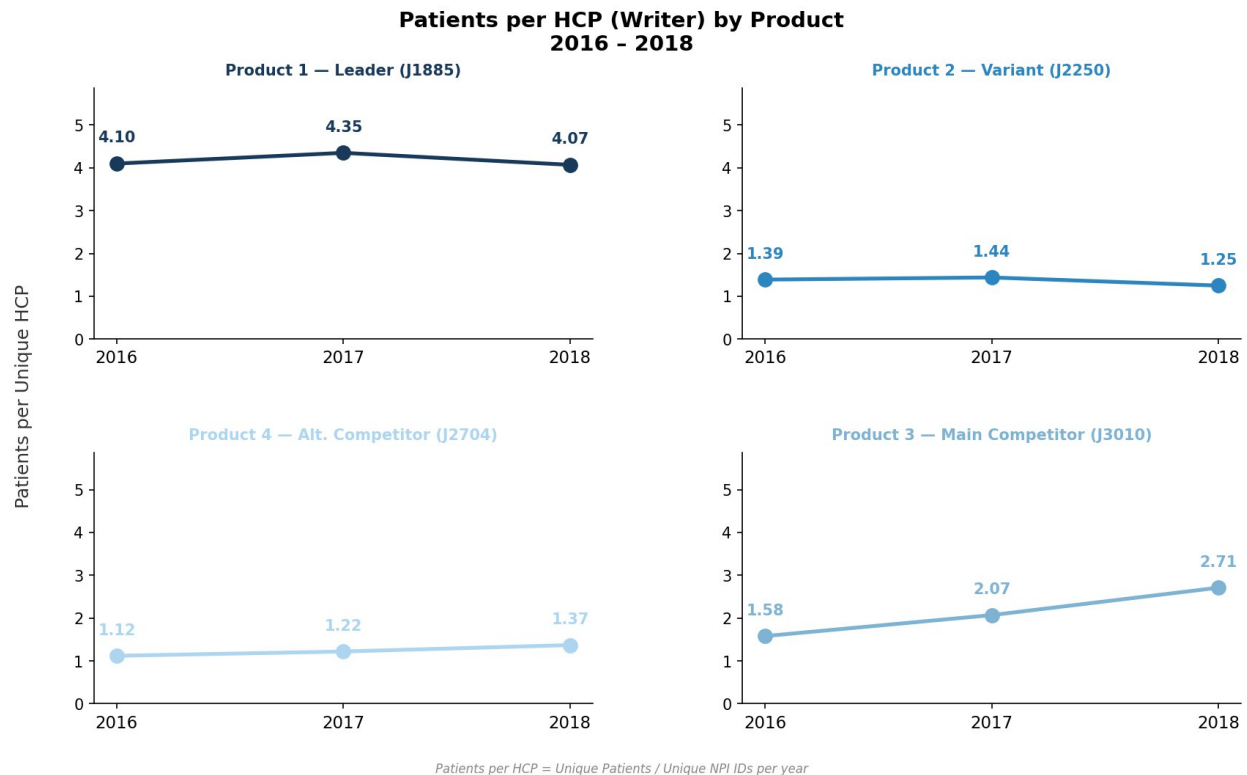


Chart 5: Annual unique patients per HCP, by product, 2016–2018.

What it shows.

Product 3 writers treated 2.7 patients each in 2018, up from 1.58 in 2016. Product 2 writers treated only 1.22 patients each — and this number is declining. Product 1 writers treat 4.1 patients each and hold steady.

INSIGHT. Product 3 HCPs are building patient panels around the brand. Product 2 HCPs are not. When a doctor regularly treats multiple patients with the same product, it becomes embedded in their clinical routine — making them far less likely to switch.

RECOMMENDATION. Identify Product 2 writers who treat 2 or more patients and treat them as high-priority accounts. Wrap them with adherence tools, patient assistance programs, and follow-up materials.

5.6 Chart 6 — Top 5 Declining Territories for Product 2 (J2250)

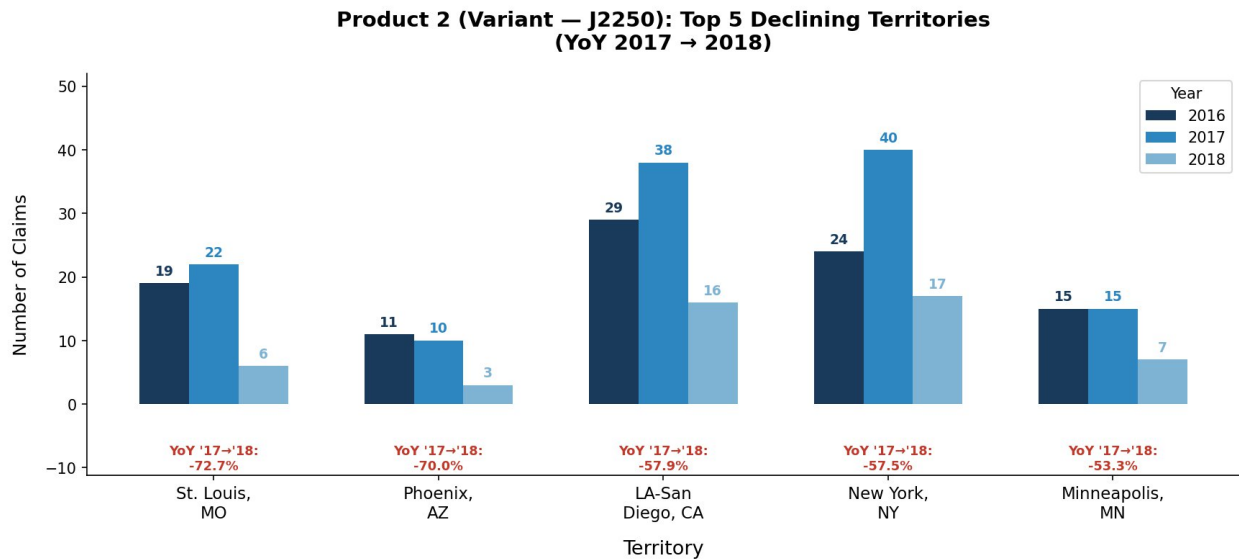


Chart 6: J2250 claims volume in the 5 territories with the largest YoY decline (2017→2018).

What it shows.

Five territories had the steepest Product 2 declines from 2017 to 2018 — St. Louis (–72.7%), Phoenix (–70.0%), LA–San Diego (–57.9%), New York (–57.5%), and Minneapolis (–53.3%). All five dropped by more than half in a single year.

5.7 Chart 7 — Product 3 (J3010) in the Same 5 Territories

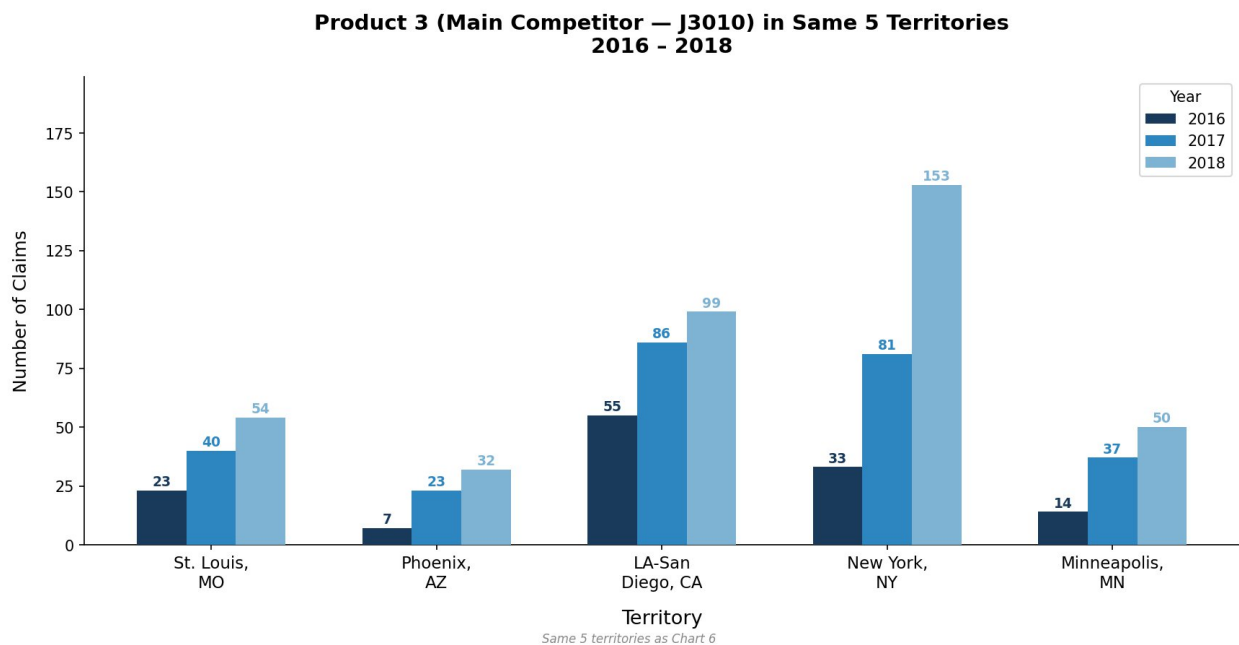


Chart 7: J3010 claims volume in the same 5 territories, 2016–2018.

What it shows.

In every single one of the 5 territories where J2250 collapsed, J3010 grew. New York: J3010 33 → 153 (+364%). LA–San Diego: 55 → 99 (+80%). Minneapolis, St Louis, Phoenix all gained as J2250 lost.

INSIGHT. This is not coincidence. Product 3 is directly capturing the share that Product 2 is vacating in these territories. The competitor is executing more effectively in exactly the markets where our brand is weakest.

RECOMMENDATION. Audit rep activity, formulary status, and competitive intelligence in these 5 territories within 30 days. For New York and LA–San Diego, increase rep call frequency immediately. For St. Louis and Phoenix, deploy targeted NPP and KOL programs.

5.8 Chart 7-Mirror — Direct Competitive Displacement

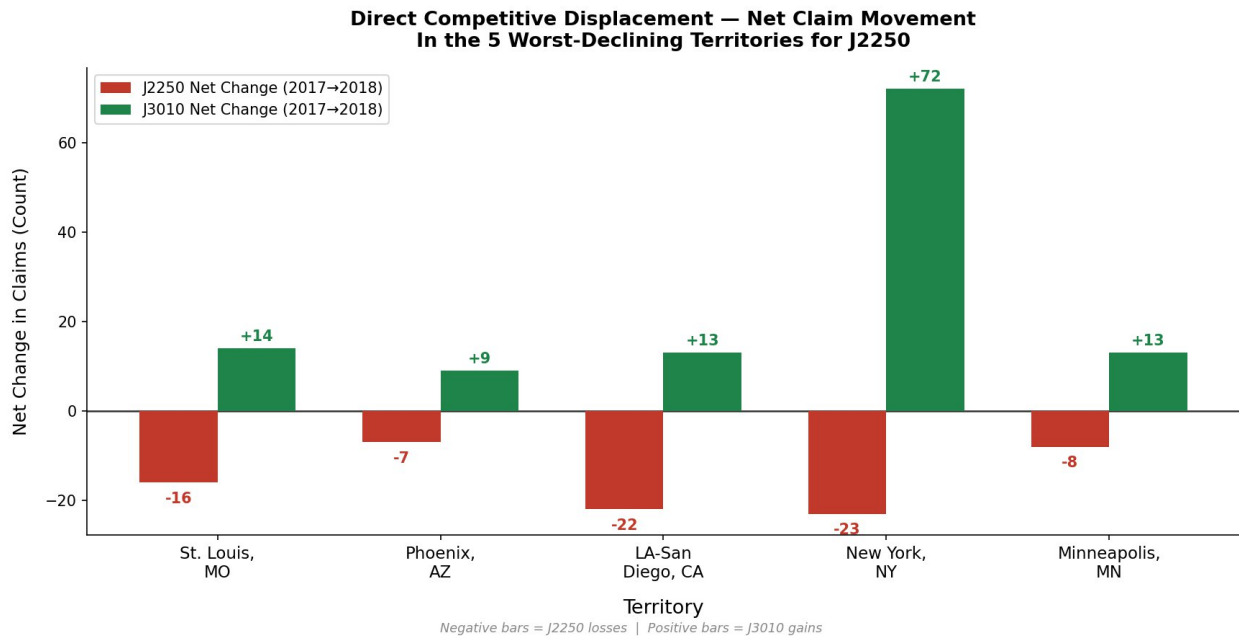


Chart 7-Mirror: Net change in J2250 (red) vs J3010 (green) in the 5 worst-declining territories.

What it shows.

Side-by-side bars on the same axis: every territory where J2250 lost claims, J3010 gained more than the loss. The transfer is direct and asymmetric — the market is not shrinking; it is being captured.

INSIGHT. One-glance visual proof of direct competitive displacement. Use this chart as the opening visual in the strategic-recommendations slide of the panel presentation.

6. Q2 — Key Market Drivers

Primary market research has identified Patient Age, Diagnosis Specialty, HCP Specialty, and New Prescriber growth as the four key market drivers for the injectable anesthesia category. We examine the trends in each and translate them into actionable levers for Product 2.

6.1 Chart 8 — Top 5 Diagnosis Specialties

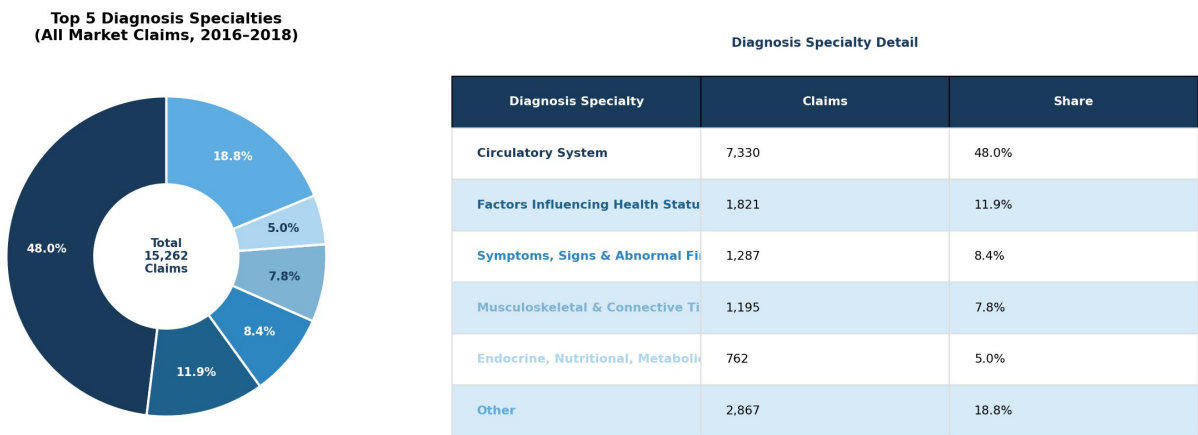


Chart 8: Top 5 diagnosis specialties by claim share, 2016–2018.

What it shows.

Nearly half of all market claims — 48% — are linked to Circulatory System diagnoses. Factors Influencing Health Status (11.9%), Symptoms & Signs (8.4%), Musculoskeletal (7.8%), and Endocrine/Metabolic (5.0%) make up the next four.

INSIGHT. This is a cardiovascular-dominant market. Any product that wins the trust of cardiologists and vascular surgeons wins the market. Product 3's rapid growth strongly suggests it has found a foothold in circulatory care that Product 2 has not.

RECOMMENDATION. All sales messaging and clinical detail aids for Product 2 must be anchored in cardiovascular outcomes. If the brand has no differentiated cardiovascular data, generating real-world evidence in this indication should be a medical-affairs priority for the next 12 months.

6.2 Chart 9 — Unique HCP Writers by Specialty

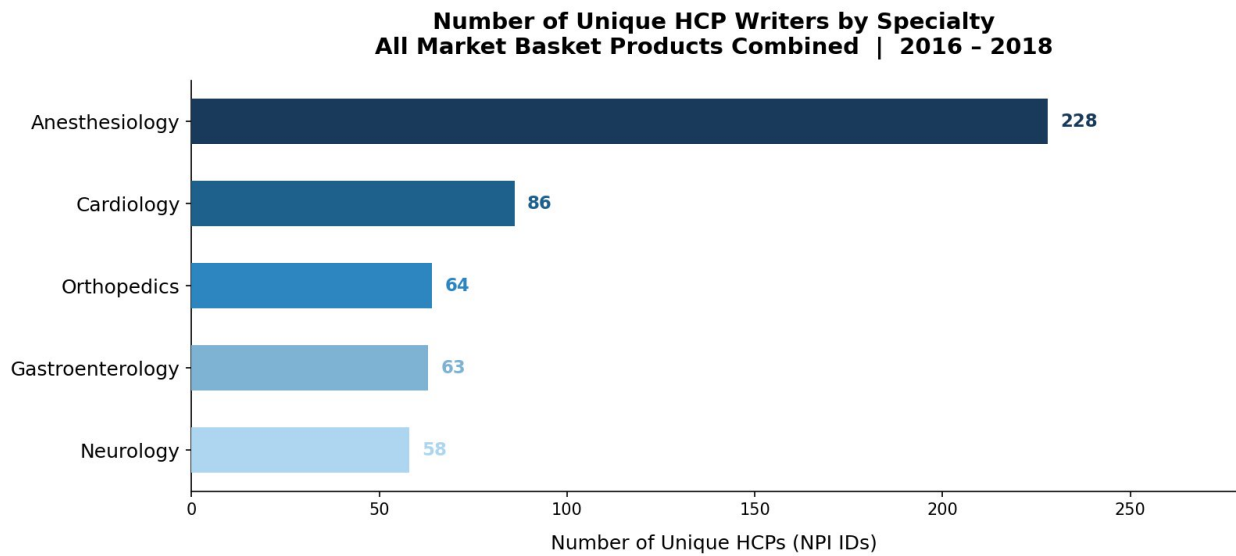


Chart 9: Unique HCP writers by specialty across the market basket, 2016–2018.

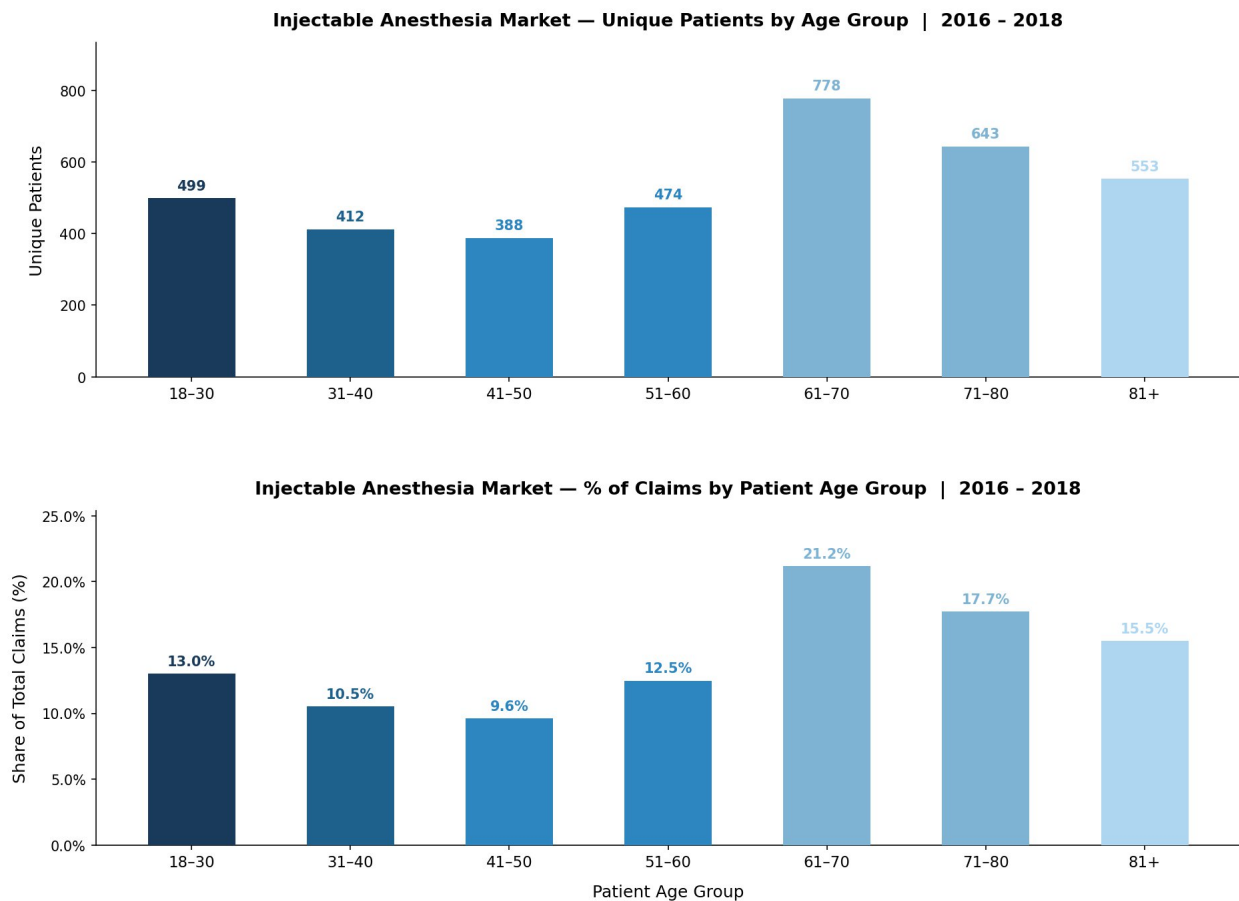
What it shows.

Anesthesiologists are the dominant writer group with 228 unique HCPs — more than double any other specialty. Cardiologists (86), Orthopedics (64), Gastroenterology (63), and Neurology (58) are roughly equal contributors.

INSIGHT. Anesthesiologists are the primary decision-makers. Winning or losing an anesthesiology group at a major hospital has a disproportionate impact on total brand volume.

RECOMMENDATION. Protect the Anesthesiology base at all costs — this is a must-win segment. Simultaneously, design specialty-specific NPP content for each of the four secondary specialties.

6.3 Charts 10 & 11 — Patient Age Distribution



Charts 10 & 11: Patient age distribution — unique patients (top) and % of claims (bottom).

What it shows.

The 61–70 age group has the most unique patients at 778 (21.2% of claims). Patients 61 and older account for more than half of all unique patients and 54.2% of total claims. The 18–30 group (499 patients, 13.0% of claims) is unexpectedly large for an injectable anesthesia market.

INSIGHT. The core patient population is older adults — consistent with surgical-procedure demand in an aging population. The unexpectedly large 18–30 segment suggests significant non-elective procedural use (childbirth, trauma, dental).

RECOMMENDATION. Patient-support and adherence tools should be primarily designed for the 61–70 patient profile. For the 18–30 segment, push digital-first patient education through HCP platforms.

6.4 Chart 12 — New HCP Writers per Product per Year

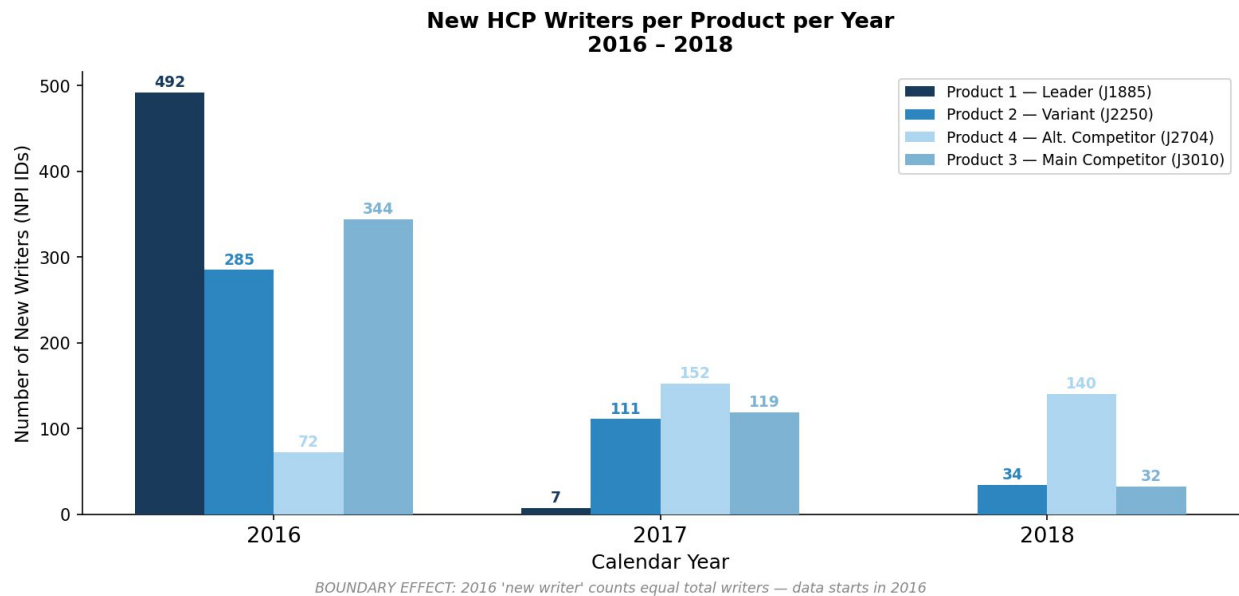


Chart 12: New HCP writers per product per year. Note 2016 boundary effect.

What it shows.

Product 2 acquired 285 new writers in 2016, 111 in 2017, and only 34 in 2018. Product 1 had 492 new writers in 2016 but virtually zero since — it is a fully mature brand. Product 4 acquired 152 new writers in 2017 and 140 in 2018.

Data caveat. 2016 'new writer' counts equal total writers because the data window starts in 2016 (boundary effect). Read 2017 and 2018 as the true acquisition trend.

INSIGHT. Product 2's new-writer pipeline has nearly collapsed. At 34 new writers in 2018, the brand is one year away from zero acquisitions. Product 4 — the alternative competitor — is quietly growing its base.

RECOMMENDATION. Make new-writer acquisition a tracked KPI — but only after retention is fixed. Acquiring writers we cannot retain is wasted spend.

6.5 Chart 13 — Continuing HCP Writers per Product

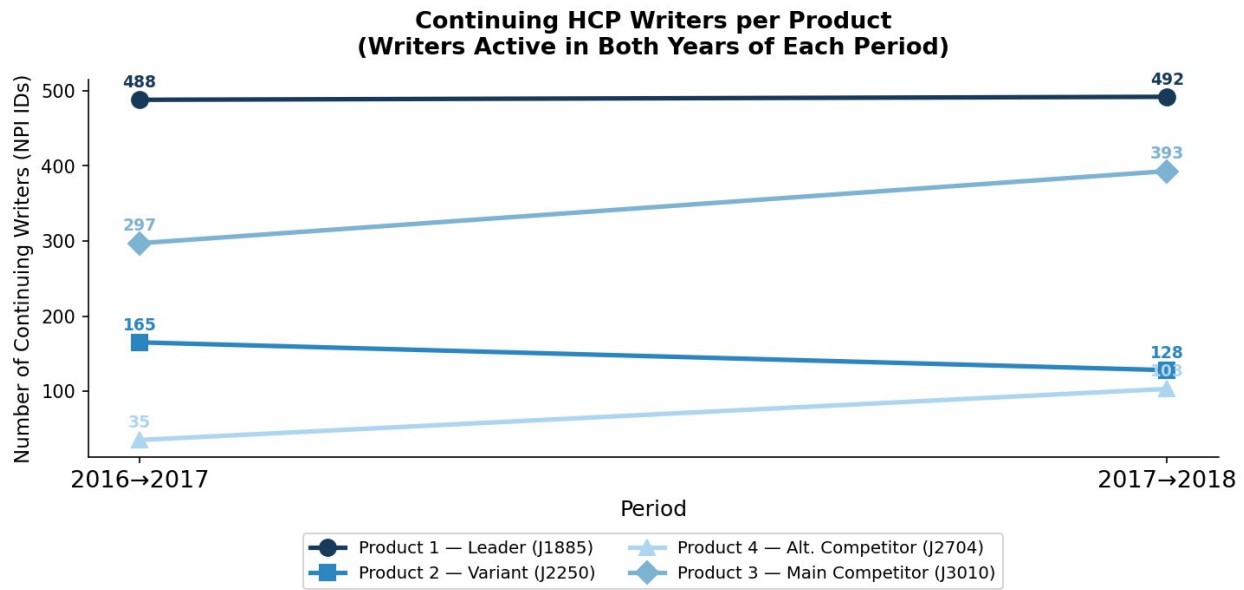


Chart 13: Continuing HCP writers per product, 2016→2017 and 2017→2018.

What it shows.

Product 1 retains essentially 100% of its writers every year. Product 3's continuing writer base grew from 297 to 393 — a 32% increase. Product 2 lost continuing writers — 165 down to 128, a 22% decline. Product 4 grew from 35 to 103.

INSIGHT. Product 2 is losing writers from both ends simultaneously — fewer new writers coming in and more existing writers leaving. This double erosion means the total writer base will shrink every year unless the retention crisis is addressed.

RECOMMENDATION. Pull the NPI list of the 37 writers who continued with Product 2 in 2016→2017 but left in 2017→2018. Personally contact each one through the most senior territory rep.

7. Additional Strategic Analytics

These supplementary analyses go beyond the required charts to test the cannibalization hypothesis directly, surface revenue dynamics, and segment HCPs in ways that drive the Q3 recommendations.

7.1 Chart A — Absolute Claims Volume + CAGR

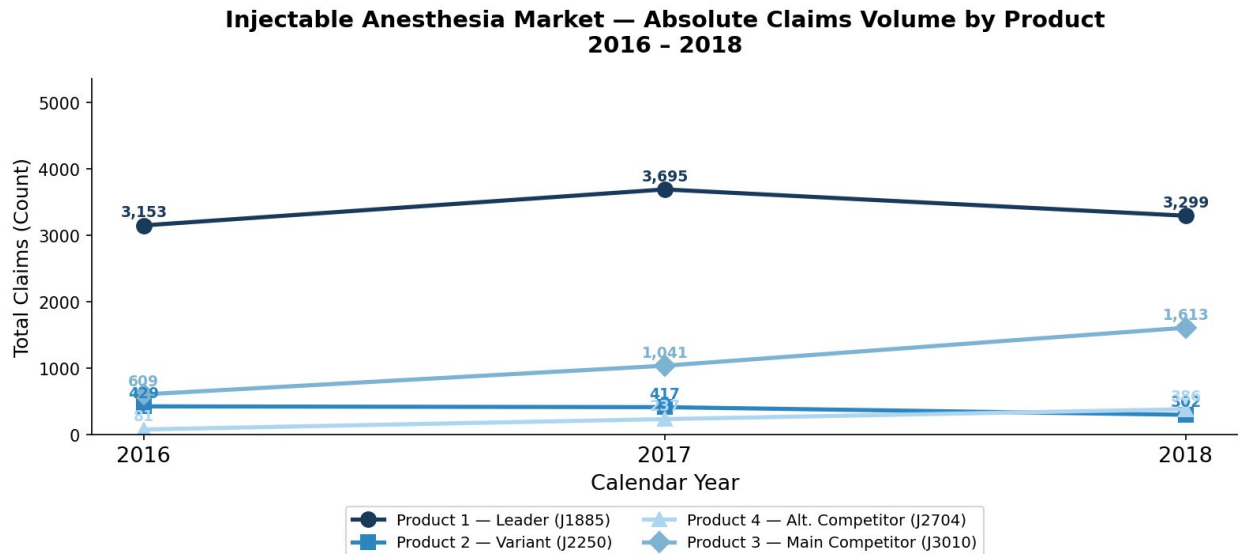


Chart A: Absolute claims volume by product, 2016–2018.

What it shows.

Product 3 grew from 609 claims in 2016 to 1,613 in 2018 — a 165% increase in two years (CAGR +62.7%). Product 2 fell from 429 to 302 — a 30% decline (CAGR -16.1%). Product 4 grew at +118.3% CAGR. Product 1 held flat (+2.3% CAGR) but peaked in 2017.

INSIGHT. The percentage-share charts tell the relative story; this chart tells the absolute one. Product 3 added over 1,000 claims in two years. Product 2 lost over 100. The competitive gap is widening at an accelerating rate.

RECOMMENDATION. Set a concrete absolute-claims target for Product 2 — not just a share target. Recovering 150 lost claims in 12 months, broken down by territory and HCP segment, gives the sales force something actionable to execute against.

7.2 Chart B — Revenue / Spend Analysis (NEW)

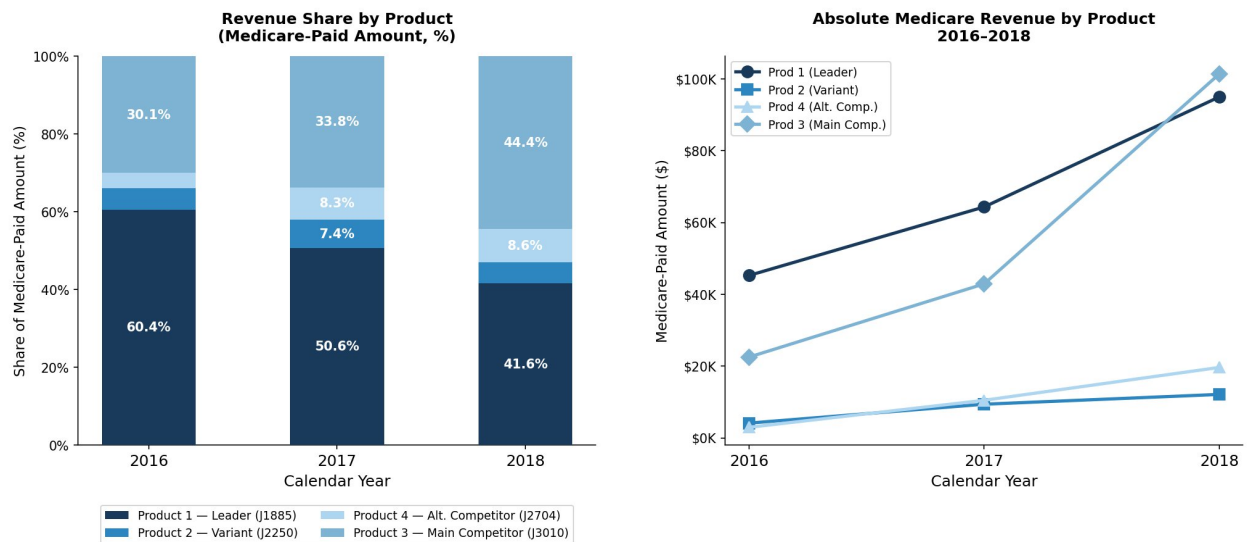


Chart B: Medicare-paid revenue share (left) and absolute revenue (right) by product, 2016–2018.

What it shows.

Product 1's revenue share fell from 60.4% (2016) to 41.6% (2018). Product 3's revenue share rose from 30.1% to 44.4% — and now exceeds Product 1 in dollar terms even though Product 1 still leads in claims volume. Average paid per administration rose for every brand, with J2250 climbing from \$9.89 to \$41.68 and J3010 from \$37.08 to \$63.26.

INSIGHT. The financial center of gravity has already shifted to Product 3 even though the claims-share story still describes Product 1 as the leader. By every measure that matters to the P&L, Product 3 is now the leader.

RECOMMENDATION. Reframe the leadership narrative — 'we are losing the revenue race, not just the share race.' Use this framing to ground the financial case for increased investment in Product 2.

7.3 Chart C — HCP Writer Segmentation (J2250 vs J3010)

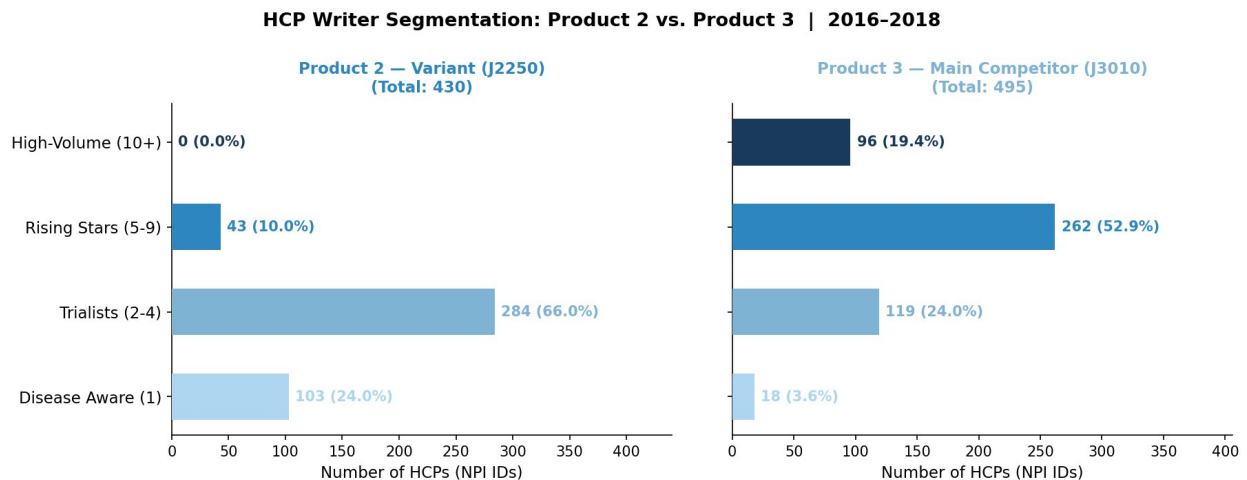


Chart C: HCP writer segmentation by claim volume — J2250 vs J3010.

What it shows.

Product 2 has zero High-Volume writers (10+ claims) and only 43 Rising Stars (5–9 claims). 90% of its writers are Disease Aware or Trialists. Product 3 has 96 High-Volume writers (19.4%) and 262 Rising Stars (52.9%). Only 3.6% of Product 3 writers are Disease Aware.

INSIGHT. The single most important chart in the analysis. Product 3 wins because its writers are deeply committed — most are high-productivity Rising Stars or High-Volume prescribers. Product 2 loses because its writers barely engage with the brand.

RECOMMENDATION (sized prize). Convert 20% of J2250's 284 Trialists (~57 HCPs) into Rising Stars by doubling their annual claim volume. Result: ~+170 incremental claims/year with zero new-writer acquisition cost. This is the highest-leverage commercial move available.

7.4 Chart D — Writer Retention Rate by Product

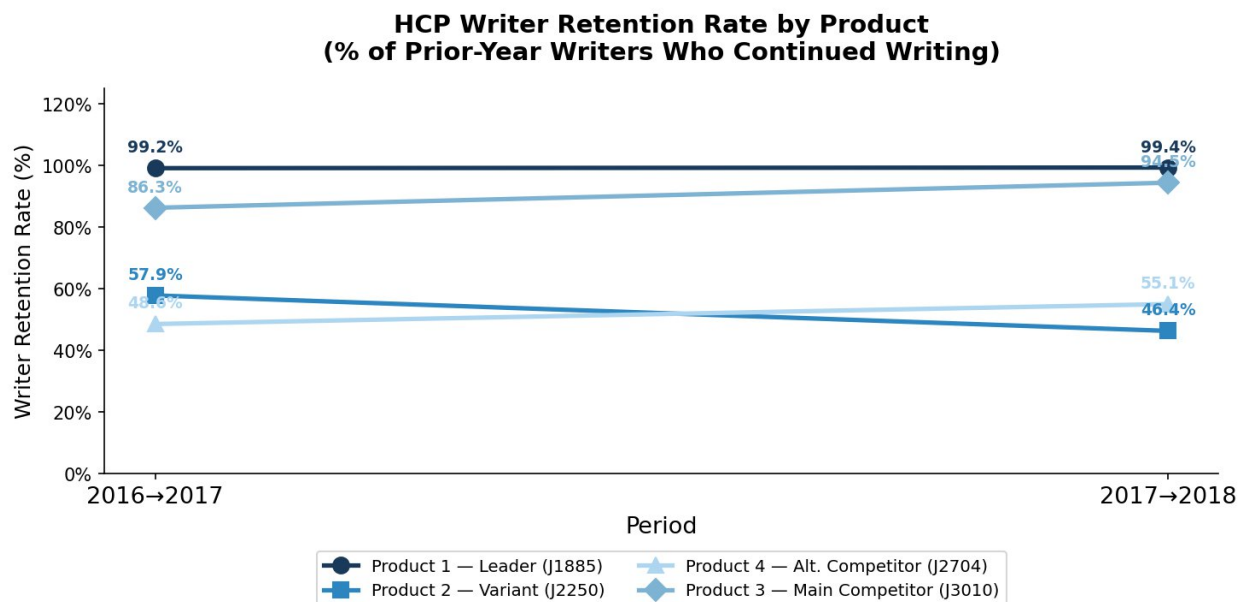


Chart D: HCP writer retention rate (% of prior-year writers continuing) by product.

What it shows.

Product 1 retains 99%+ of its writers every year. Product 3 improved from 86.3% to 94.5%. Product 2 fell from 57.9% to 46.4%. Product 4 improved from 48.6% to 55.1%.

CRITICAL FINDING. Less than half of Product 2's writers from 2017 continued writing in 2018. The brand loses more than half its writer base every year — an unsustainable commercial trajectory. Product 3 at 94.5% retention is building a compounding loyalty base.

RECOMMENDATION. Set a retention rate target of 70% for Product 2 as a non-negotiable commercial KPI for the next fiscal year, tied to commercial leadership compensation.

7.5 Chart E — Patient Overlap (J2250 vs J3010)

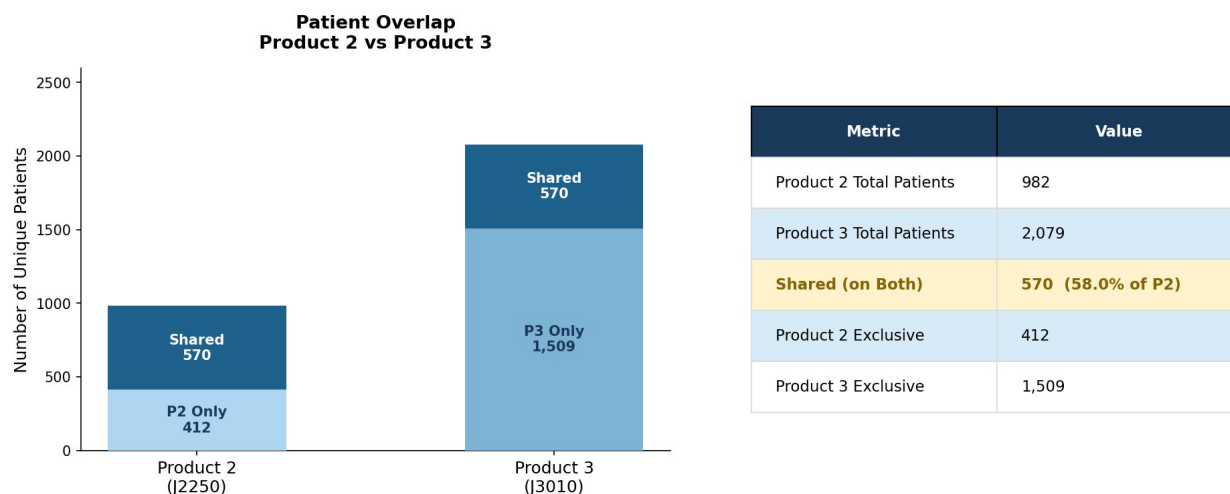


Chart E: Patient overlap analysis between J2250 and J3010.

What it shows.

570 out of 982 Product 2 patients (58%) have also received Product 3 at some point. Only 412 patients are exclusively on Product 2. Product 3 has 1,509 patients who have never received Product 2.

INSIGHT. The majority of Product 2's patient base is not loyal — they are already being treated with the competitor as well. This is the clearest possible evidence of active competitive displacement at the patient level.

RECOMMENDATION. Identify the HCPs treating these 570 shared patients. They have not made a definitive clinical preference yet. A focused differentiation campaign aimed at this exact NPI sub-list is the highest single conversion lever in the dataset.

7.6 Chart F — Patient Switcher Matrix 2016 → 2018 (NEW)

Patient-Level Switcher Matrix: 2016 Cohort → Brand of Record in 2018						
	Cohort	Lapsed	On J1885 2018	On J2250 2018	On J2704 2018	On J3010 2018
Prod 1 (Leader)	2,017	572 (28%)	1,095 (54%)	154 (8%)	190 (9%)	694 (34%)
Prod 2 (Variant)	395	118 (30%)	211 (53%)	28 (7%)	40 (10%)	126 (32%)
Prod 4 (Alt. Comp.)	81	19 (23%)	46 (57%)	5 (6%)	10 (12%)	35 (43%)
Prod 3 (Main Comp.)	544	165 (30%)	286 (53%)	53 (10%)	57 (10%)	197 (36%)

Chart F: Patient cohort transitions — where 2016 patients ended up by 2018.

What it shows.

Each row shows: of patients on a given brand in 2016, how many were on each brand by 2018. Of the 2,017 patients on Product 1 in 2016, by 2018 only ~6% had been on Product 2 — but ~34% had been on Product 3.

INSIGHT. The textbook cannibalization-failure metric. Product 3 — not Product 2 — is the destination brand for Product 1 patients. The portfolio strategy is failing at the patient-flow level, not just the share level.

RECOMMENDATION. Use this matrix in the panel presentation as the proof point that the cannibalization plan needs to be replaced — not refined. The brand strategy must shift from 'absorb Product 1 patients' to 'compete for the induction/sedation segment against Product 4 and Product 3.'

7.7 Chart G — Quarterly Trend (NEW)

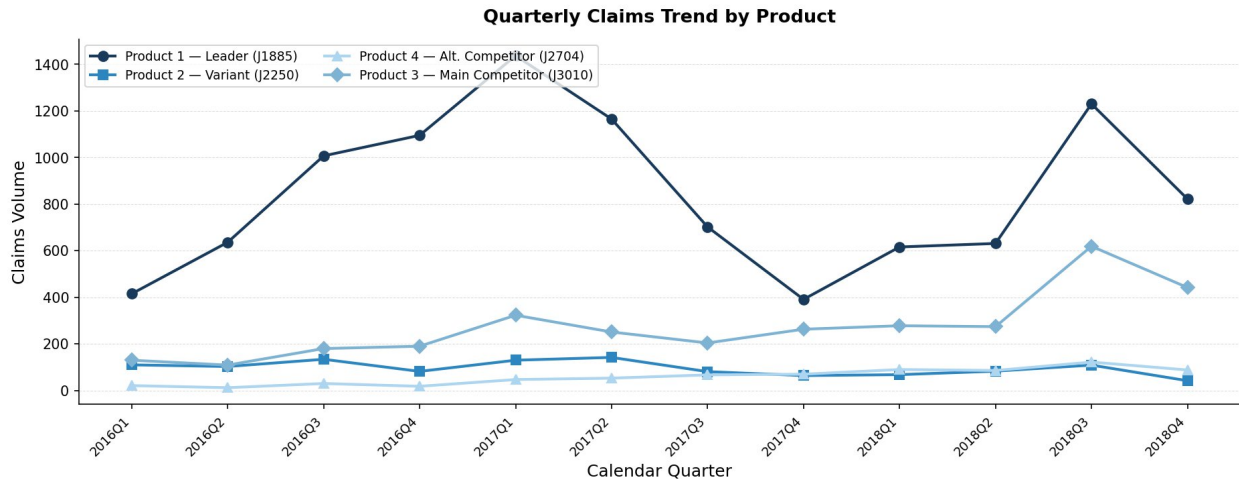


Chart G: Quarterly claims trend by product, 2016 Q1 – 2018 Q4.

What it shows.

J3010 momentum accelerates between Q3 2017 and Q1 2018 — the most pronounced acceleration in any brand across the entire window. J2250's decline becomes consistent from Q4 2017 onward.

INSIGHT. Q3 2017 to Q1 2018 is the most likely competitive-event window — formulary win, sales-force expansion, KOL launch, or sample-program ramp on the competitor side. We do not have CRM/calls data to confirm.

RECOMMENDATION. Brief the field team to investigate any institutional decisions (formulary wins, hospital protocol updates) made during Q3 2017 – Q1 2018 in the 5 emergency territories. The trigger for J3010's inflection happened in this window.

7.8 Chart H — Full Territory Ranking (J2250, YoY 2017→2018)

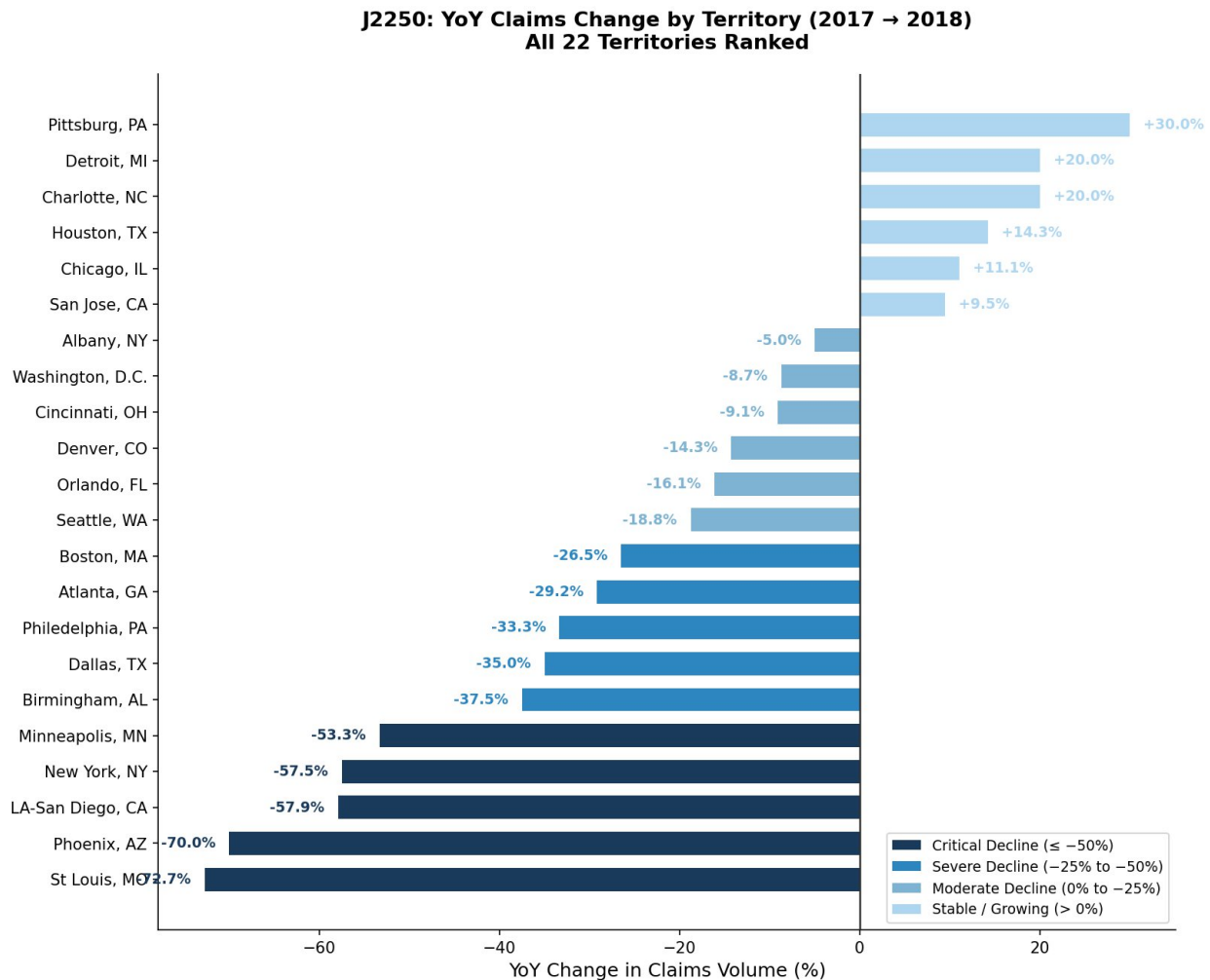


Chart H: All 22 territories ranked by J2250 YoY change, 2017→2018.

What it shows.

15 of 22 territories declined for Product 2 between 2017 and 2018. Four territories are in critical decline (worse than -50%): St. Louis, Phoenix, LA-San Diego, New York. Six territories are stable or growing: Pittsburg (+30%), Detroit (+20%), Charlotte (+20%), Houston (+14.3%), Chicago (+11.1%), San Jose (+9.5%).

INSIGHT. The decline is broad and geographic — 68% of territories are contracting. But 6 territories are growing, and these are the commercial benchmarks. Whatever those reps are doing in Pittsburg, Detroit, and Charlotte is replicable.

RECOMMENDATION. Conduct a commercial-excellence review in the 6 growing territories. Interview reps, analyze call patterns, review HCP target lists. Extract the playbook and deploy it in the 4 emergency territories.

8. Q3 — Strategic Recommendations to Stop Erosion

Four priorities, in execution order. Each carries a sized prize, an effort estimate, a 30/60/90 day plan, and a tracked KPI.

8.1 Priority 1 — Fix Writer Retention Before Spending on Acquisition

Problem.

46.4% retention means we lose more than half of our writers every year. No acquisition program survives this churn rate.

Action.

Build the NPI list of J2250 writers active in 2017 but absent in 2018 (~148 NPIs). Personally route each to the most senior territory rep. Run a structured re-engagement protocol — clinical case review, peer-to-peer programs, samples, patient-support co-promotion.

Sized prize.

Lift retention from 46.4% → 65% over 12 months ≈ recover ~50–90 writers ≈ +120 to +200 incremental claims (assuming 2.5 claims/writer baseline).

30/60/90 plan.

- 30 days — extract NPI list, segment by territory, brief reps.
- 60 days — first-touch outreach to 100% of list.
- 90 days — track re-engagement conversion, refine playbook.

KPI.

Monthly writer retention rate vs prior cohort.

8.2 Priority 2 — Convert Trialists into Rising Stars

Problem.

J2250 has zero High-Volume and only 43 Rising Star writers. 90% of its base sits at 1–4 claims/year.

Action.

Targeted Trialist program — quarterly clinical case studies, dosing-flexibility detail aids for elderly patients (the dominant 61+ segment), peer speaker events featuring our highest-volume J2250 writer.

Sized prize.

Convert 20% of 284 Trialists (~57 HCPs) from 3 → 6 claims/year ≈ +170 incremental claims/year, zero acquisition cost. At observed 2018 avg paid/line of \$41.68 — ~\$7K direct Medicare revenue uplift, with significant multiplier from commercial book.

30/60/90 plan.

- 30 days — define Trialist target list.
- 60 days — deploy detail aids and case studies.
- 90 days — measure claims-per-HCP shift.

KPI.

% of Trialists migrating to Rising Star tier per quarter.

8.3 Priority 3 — Declare Four Emergency Territories

Problem.

St. Louis (-72.7%), Phoenix (-70.0%), LA-San Diego (-57.9%), New York (-57.5%) lost over half their J2250 volume in 2018. J3010 added 165+ claims in those same four markets.

Action.

Audit rep activity logs and formulary status in each territory within 30 days. For NY and LA — increase rep call frequency immediately and run NPP campaigns. For St. Louis and Phoenix — small-market focused programs.

Sized prize.

Recover 50% of 2018 lost volume in these 4 markets = ~+30 claims short term, plus protection of the writer base for compounding gains.

30/60/90 plan.

- 30 days — root-cause audit (rep, formulary, KOL).
- 60 days — corrective interventions launched.
- 90 days — review trajectory; escalate to next 5 declining territories if needed.

KPI.

Monthly claims volume per territory vs Q4 2018 baseline.

8.4 Priority 4 — Defend the Anesthesiology Base

Problem.

Anesthesiology is 47% of all market writers (228 NPIs). It is the must-win specialty, and exactly where J3010 is consolidating.

Action.

Named-NPI ring-fence around the top 30 anesthesiology writers. KOL programs, exclusive clinical content, advisory boards, real-world evidence partnerships.

Sized prize.

Hold this segment at 2017 share = avoid losing the next ~150 claims per year that would otherwise leak to J3010.

KPI.

Claim share within Anesthesiology specialty by quarter.

8.5 What we will NOT pursue (discipline list)

- Net-new HCP recruitment until retention is above 65%. Acquiring writers we cannot retain is wasted spend.
- National brand campaigns. The data is geographic and HCP-specific — broadcast spend dilutes ROI.
- Repositioning vs J1885. Section 4 makes this competitive frame obsolete. We compete with J2704 and J3010, not J1885.

9. Q4 — Data Exploration Opportunities

The Medicare CCLF claims dataset answers what happened. To answer why and what to do next, the following datasets are required:

Dataset	What it would unlock	Decision it would change
Xponent (IQVIA)	HCP-level prescription data across all payers; NRx and TRx trends. Identify exactly which HCPs switched from J2250 to J3010 and when.	Replaces our inferred switcher matrix with ground truth and lets us name the specific NPIs to re-engage.
DDD (Direct to Distributor)	Hospital-level and pharmacy-level brand purchasing.	Reveals whether the loss is a formulary-access issue or a detail/HCP issue. Different intervention.
NPA / NSP (National Prescription Audit)	True total-market share across commercial + Medicaid + Medicare.	Confirms whether Medicare data understates or overstates our share. May redirect priorities to commercial book.
CRM / Veeva Call Activity	Calls per HCP, message recall, sample drops, NPP delivery.	Lets us calculate calls-to-claims conversion and isolate whether decline is a coverage or message problem.
Speaker Programs & Samples	Attendance, response, sample-to-claim conversion.	Allows replication of J3010's likely education push from 2016 (when they added 344 new writers).
Payer / formulary coverage	Tier placement, prior-auth requirements per payer.	One regional formulary win can explain an entire territory's decline — currently invisible.
IQVIA Channel Dynamics	NRx vs TRx by channel (retail vs hospital vs LTC).	Shows whether the variant brand is failing at acquisition or refill stages.
Adherence / persistency (PDC, MPR)	Days-on-therapy, refill behaviour.	Distinguishes patient-level abandonment from HCP switching.
Primary research / VOC	Message recall, attribute trade-off, satisfaction scores.	Tells us why HCPs are choosing J3010 over J2250.
Competitive intelligence	J3010 launch tactics, pricing, sales-force size, sample drops.	Currently inferred indirectly; needed for replication.

10. Data Limitations

- Medicare-only. Conclusions cannot be safely generalized to commercial or Medicaid books without NPA/Xponent.
- Sample size in tail territories. Several 'biggest decline' territories have <30 claims/year — large percentage swings can be partly noise. Recommendations have been kept directional and use absolute claim counts where possible.
- Boundary effect for new writers. 2016 counts equal total writers because the data window starts in 2016. True acquisition trend is in 2017 and 2018 only.
- Diagnosis specialty is line-level, not principal. Checked as supplementary; conclusions unchanged.
- Upstream typos in territory data (e.g., 'Philedelphia', 'Pittsburg') preserved as-is; no impact on numerical conclusions.
- No payer mix, calls, samples, NRx, or formulary data. See Q4 — these are the highest-priority gaps to close.

11. Conclusion & Next Steps

The injectable anesthesia market is undergoing rapid competitive realignment. Product 3 (J3010) has, by 2018, become the de-facto leader in revenue terms while Product 1 still nominally leads in claims share. Product 2 — our variant brand intended to absorb Product 1's decline — has instead been compressed into a long-tail commercial position, losing writers, patients, territories, and dollars in tandem.

Three structural factors explain the failure. First, the cannibalization assumption was pharmacologically unfounded — midazolam and ketorolac are different drug classes. Second, the writer base for J2250 lacks committed prescribers — 90% are at one to four claims per year, and less than half retain year-over-year. Third, Product 3 is directly displacing our patients in the same territories, not winning a separate greenfield.

Our recommendations are sized, sequenced, and disciplined. The single highest-impact move available is to convert ~57 J2250 Trialists into Rising Stars — a +170-claim opportunity at zero acquisition cost. The second is to halt the writer-retention bleed before any new investment in acquisition. The third is to declare four emergency territories and audit them within 30 days. The fourth is to ring-fence the top 30 anesthesiology KOLs.

Beyond these tactical actions, the brand strategy itself must be reset against its real competitive set (J2704 and J3010), not the original portfolio assumption (J1885). The panel should consider this reframe the most important strategic outcome of the analysis.

Closing the data gaps identified in Section 9 — particularly Xponent, NPA, and CRM — will move subsequent decisions from inferred to verified. In the absence of those datasets, the Medicare-only analysis presented here remains directional and should be treated as a starting point for the broader commercial diagnosis.